

Skin Wrinkle

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Figure 1: *Skin Wrinkle* applied to the Snake and Goanna characters from the DreamWorks Animation Short *Bilby*.

ABSTRACT

Skin Wrinkle is a custom deformer tool at DreamWorks Animation that produces wrinkles in the animated skin mesh geometry solely based on the incoming animated skin mesh, rest skin mesh, and artist specified parameters. This wrinkle effect on characters that exhibit dynamic folds enhances their look and animation performance. The deformer minimizes loss in body volume as the skin wrinkles and also performs continuous self collision detection and resolution to handle the tight wrinkles. *Skin Wrinkle* is time-independent, fast, robust, and controllable, thus making it very artist-friendly.

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1 MOTIVATION

Several characters such as snakes and goannas exhibit dynamic skin wrinkles as they move around [Figure 1]. Creating these skin wrinkles in the animated skin mesh enhances the look of those characters and their motion. Skin meshes from Animation typically are wrinkle free. Such animated skin can stretch or compress potentially causing distortion. Sometimes animated skin may come with static wrinkles baked in or with wrinkles in specific locations animated with pre-built rigging controls, and are not dynamic in

nature. All of this necessitates the addition of dynamic wrinkles post animation to enhance the look and motion of the characters.

Performing a dynamics simulation using a bone-muscle-fat-fascia-skin model can be expensive. It would involve several departments, including Modeling, Rigging, Animation, and Character Effects, to build and move the anatomical structures and perform that simulation. Such a simulation would also be time dependent, time intensive, difficult to control, and would have issues with characters that are exaggerated in animation.

The *Skin Wrinkle* deformer produces wrinkles solely based on the incoming animated skin mesh [Figure 2], rest skin mesh, and artist parameters without the need of a complex dynamics setup, significantly saving time and resources. While minimizing volume loss during wrinkling, it also performs continuous self collision and resolution to handle the tight wrinkles robustly. It can handle exaggerated animation by providing an animated rest mesh that captures that exaggeration. The deformer focuses on the wrinkling aspect of the simulation which is dependent on the pose of a character at a certain frame without deploying the dynamic components such as the gravity force or the character's motion. Since *Skin Wrinkle* is a time-independent deformer, it can run on any chosen time frame and also be multi-threaded across frames. The steps in the *Skin Wrinkle* process are described next in Section 2.

2 SKIN WRINKLE

2.1 Pre-processing

The input to *Skin Wrinkle* is an animated mesh $mesh_{anim}$, and a rest mesh $mesh_{rest}$. Internally, these quadrilateral meshes are first triangulated. The size of the triangles affect the resolution or fidelity of the wrinkles. For finer wrinkles, a higher resolution triangulated mesh is preferred. The artist is provided controls to affect the size of the triangles. The original input mesh is wrapped to the skin-wrinkled triangulated mesh for the final output.

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2.2 Simulation Setup

At setup, stretch springs $Spring_{stretch}$ are created at each edge connecting the two vertices of that edge. When two triangles share an edge, bend springs $Spring_{bend}$ are created connecting the two vertices of those triangles that are not part of that edge. The rest lengths of the springs are initialized to be the lengths between connected vertices at rest pose. The rest mesh is a live input which can be modified per frame if needed (for example, a rest mesh capturing the exaggerated animation). In order to minimize volume loss during wrinkling, a collision mesh, $mesh_{collision}$ is setup. Points of $mesh_{collision}$ are offset inside $mesh_{anim}$ along the inverse of its point normals using an artist specified offset distance $offset_{dist}$.

2.3 Simulation

For an animated pose, the spring lengths are initialized to be the lengths between connected vertices at that pose. A mass-spring simulation is then performed. Simulation parameters such as mass, stretch spring constant, stretch spring damping, bend spring constant, bend spring damping, global damping, and substeps can be controlled by the artist. The parameters, except substeps and global damping, can be locally scaled by setting point attributes (typically done by skin painting). With respect to *Skin Wrinkle*, the substeps corresponds to the number of sub steps it takes to go from the animated pose to the wrinkled pose. *Skin Wrinkle* focuses on the wrinkling aspect which is dependent on the current pose of the animated mesh in relation to its rest pose. *Skin Wrinkle* does not deploy external dynamic components such as the gravity force. The velocity of the points are set to zero at the start of any frame of simulation to remove the influence of the character's motion during the wrinkle focused simulation. Along with the simulation parameters, the size of the mesh triangles affect the wrinkle fidelity.

2.4 Minimizing Volume Loss

Performing a skin wrinkle simulation just on the animated skin mesh surface can cause it to lose volume or collapse/buckle into the volume of the body. In order to avoid that and minimize volume loss during the wrinkle simulation, an internally offset mesh $mesh_{collision}$, as described in Section 2.2, is used as a collision object. The artist can tune the $offset_{dist}$ used to create the collision object to obtain the desired visual volume maintenance.

2.5 Self Collision

Since the wrinkles can be tight as the character moves around, self collision becomes important to obtain a realistic result. *Skin Wrinkle* employs continuous self collision detection and resolution. Both vertex-to-triangle collision and edge-to-edge collision are taken into account. Before the continuous self collision resolution, the mesh can interpenetrate itself after simulation, especially for larger substeps. After the mass-spring simulation on a frame, for continuous collision detection, the exact moment $moment_{collide}$ at which the self collision would have happened is first calculated. Vertices are then backtracked to $moment_{collide}$ and their positions then, which are collision free, are set to be the final positions for that frame. Additionally, in order to represent skin thickness, a self collision offset parameter, $selfcollision_{offset}$, is provided which can be minimally used to offset the skin further during backtracking.

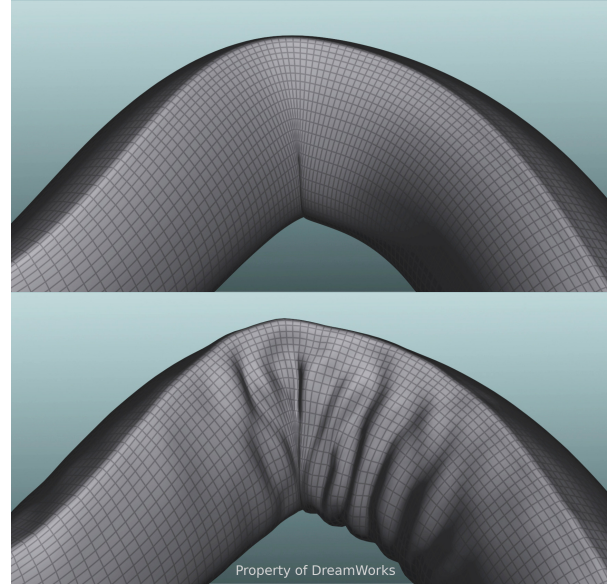


Figure 2: Comparison of pure animation mesh (top) vs mesh with wrinkles created using *Skin Wrinkle* (bottom)

3 IMPLEMENTATION

Skin Wrinkle is implemented as a time-independent deformer using custom nodes in a third-party package. The inputs to the tool are an animated mesh and a rest mesh. The tool provides various parameters to vary the triangulation, control wrinkling effect using simulation parameters, minimize volume loss, and perform continuous self collision to represent the tight skin wrinkles. The artist is also provided a post blend control that can be used to locally blend regions back to the original animation based on artistic direction as needed, such as not to affect the facial performance.

4 RESULTS AND CONCLUSION

Skin Wrinkle is used by Character Effects artists at DreamWorks. For a mesh with 30,000 points *Skin Wrinkle* runs at ≈ 3 frames per second. Since it is a time-independent deformer, a shot could be multi-threaded across frames. Since the deformer operates on the animated input mesh and is consistent in its execution, the wrinkles maintain the frame-to-frame coherency that occurs in the animated input. While the *Skin Slide* tool [Somasundaram et al. 2023] can be used to provide sliding motion to represent the characters' weight, the *Skin Wrinkle* deformer can be used to represent the nature of the characters' skin. For characters that need both effects, the *Skin Wrinkle* deformer can be run after the *Skin Slide* tool to represent both the characters' weight and skin's nature. Wrinkles on skin are more prominently visible when thick fur does not cover the skin. *Skin Wrinkle* adds more realism to the characters, and enhances their screen presence and animation performance.

REFERENCES

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